

On the Threshold of Having a Linear Treewidth in Random Graphs

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Abstract. The concept of tree-width and tree-decomposition of a graph plays an important role in algorithms and graph theory. Many NP-hard problems have been shown to be polynomially solvable when restricted to the class of instances with a bounded tree-width. In this paper, we establish an improved lower bound on the threshold for a random graph to have a linear treewidth, which improves the previous result by Klops [1].

1 Introduction

The concept of tree-width and tree-decomposition of a graph plays an important role in the study of algorithm and graph theory [2,1]. Many NP-hard problems have been shown to be polynomially solvable when restricted to the class of instances with a bounded tree-width. Dynamic programming algorithms based on the tree-decomposition of graphs have found many applications in research field such as artificial intelligence[3,4].

Over the past ten years, there has been much interest in the study of the phase transitions and typical-case behavior of problem instances randomly-generated from some probability distribution [5,6,7,8]. Randomly-generated problem instances have been widely used as benchmarks to test the efficiency of algorithms and to gain insight on the problem hardness.

The theory of random graphs pioneered by the work of Erdős [9] deals with the phase transitions and threshold phenomena of various graph properties such as the connectivity, the colorability, and the size of (connected) components. In [1], Klops studied the threshold phenomenon for a random graph to have a treewidth linear to the number of vertices, and proved that **whp** a random graph $G(n, m)$ with $\frac{m}{n} > 1.18$ has a treewidth linear in n . Klops commented that it was not known whether his lower bound 1.18 can be further improved and that the treewidth of a random graph $G(n, m)$ with $\frac{1}{2} < \frac{m}{n} < 1$ is unknown [1]. In [10], it is shown that the threshold phenomena of having a linear treewidth can be used to explain the results of many empirical studies in AI. To the best knowledge of the author, no further result has been obtained regarding the lower bound on the treewidth of random graphs since Klops' work.

The purpose of this paper is to establish a lower bound on the threshold for a random graph to have a linear treewidth, which improves the previous result by Kloks.

2 Preliminary and the Main Result

The *treewidth* can be defined in several equivalent ways. The one that is the easiest to state is via the *k-tree* defined recursively as follows ([1]):

1. A clique with $k+1$ vertices is a k -tree;
2. Given a k -tree T_n with n vertices, a k -tree with $n+1$ vertices is constructed by adding to T_n a new vertex and connecting it to a k -clique of T_n .

A graph is called a partial k -tree if it is a subgraph of a k -tree. The treewidth $tw(G)$ of a graph G is the minimum value k for which G is a partial k -tree.

Let $\mathcal{P} = (\Omega, \mathcal{A}, \Pr)$ be a probability space where Ω is a sample space, $\mathcal{A}$ is a σ -field, and $\Pr$ is a probability measure. Throughout this paper, we will use the following notations:

- $\mathcal{E}_{\mathcal{P}}[X]$: the expectation of a random variable X ;
- $\sigma_{\mathcal{P}}^2[X]$: the variance of a random variable X ;
- I_A : the indicator function of an event $A \in \mathcal{A}$.

When the probability space is clear from the context, we will suppress the subscripts and simply write $\mathcal{E}[X]$, $\sigma^2[X]$, and I_A .

Let $\{\mathcal{P}_n = (\Omega_n, \mathcal{A}_n, \Pr_n), n \geq 1\}$ be a sequence of probability spaces and let $\{A_n \in \mathcal{A}_n, n \geq 1\}$ be a sequence of events. We say that $\{A_n \in \mathcal{A}_n, n \geq 1\}$ occur *with high probability* (**whp**) if $\lim_n \Pr_n\{A_n\} = 1$.

A random graph $G(n, m)$ over n vertices is defined to be a graph with m edges selected uniformly and randomly without replacement from all the $\binom{n}{2}$ possible edges [11]. Kloks proved that **whp**, a random graph $G(n, m)$ with $\frac{m}{n} > 1.18$ has a treewidth linear in n . Kloks commented that it was not known whether his lower bound 1.18 can be further improved and that the treewidth of a random graph $G(n, m)$ with $\frac{1}{2} < \frac{m}{n} < 1$ is unknown [1]. To my best knowledge, no further result has been obtained regarding the treewidth of $G(n, m)$ since Kloks' work.

Our main result is an improved lower bound on the threshold of having a linear treewidth. The improvement comes from two factors: (1) the use of a new combinatorial construct to make better use of the first moment method; and (2) the use of a random graph equivalent to $G(n, m)$ that makes it possible to have a more accurate estimation of some quantity.

Theorem 1. *For any $\frac{m}{n} = c > 1.081$, there is a constant $\delta > 0$ such that*

$$\lim_n \Pr \{ tw(G(n, m)) > \delta n \} = 1. \quad (1)$$